**Project Name: FraudX – Real-Time Fraud Detection System**

## 1. Overall Approach

To address the challenges of rising digital payment fraud, a **Real-Time Intelligent Fraud Detection Architecture** has been designed. This system focuses on **preventing fraudulent transactions before completion**, rather than detecting them afterward.

The core strategy is **Behavior-Based Risk Analysis**. By analyzing user transaction patterns (amount, device, location, frequency), the system identifies anomalies and assigns a **dynamic risk score**.

The application features distinct system layers:

- **User Interface (Frontend):** Enables transaction simulation, displays alerts, and provides explanations for flagged transactions.
- **Fraud Detection Engine:** Processes transaction data, computes risk scores, and determines actions in real time.
- **Admin Dashboard:** Provides analytics, fraud monitoring, and system control.

## 2. System Architecture Overview

The system follows a modular **3-Tier Architecture**:

**Frontend (Presentation Layer):**

- **User View:**
  Provides a transaction simulation form capturing details like amount, receiver, device, and location. Displays real-time results (Allow / OTP / Delay / Block).
- **Admin Dashboard:**
  Displays analytics such as total transactions, fraud cases, and risk distribution using charts and tables.

**Backend (Business Logic Layer):**

- Acts as the central controller for fraud detection.
- Implements:
  - **Feature Extraction Logic** (behavioral signals)

    - **Risk Scoring Engine** (rule-based + ML)
    - **Decision Engine** (adaptive response system)
- Communicates with ML service for fraud probability prediction.

**Data Persistence Layer:**

- Stores:
    - User behavioral profiles
    - Transaction logs
    - Risk scores and decisions
- Ensures data consistency and supports analytics.

**3. Key Components**

The system is composed of the following core logical modules:

**Behavior Profiling Engine**

- Maintains user-specific patterns:
    - Average transaction amount
    - Known devices
    - Frequent receivers

**Risk Scoring Engine**

- Combines:
    - Rule-based scoring (amount deviation, new device, etc.)
    - Machine Learning probability
- Generates a **final risk score** for each transaction.

**Decision Engine**

- Determines action based on risk score:
    - Low Risk → Allow
    - Medium Risk → OTP Verification
    - High Risk → Delay + Alert

    - Very High Risk → Block Transaction

**Explainable AI Module**

- Provides clear reasons for flagged transactions:
    - “High amount deviation”
    - “Unknown receiver”
    - “New device detected”

**Fraud Advisor Chatbot**

- Allows users to:
    - Ask why a transaction was blocked
    - Understand risk factors
- Enhances transparency and user trust

**4. Data Flow Explanation**

1. **Transaction Submission:**
   User enters transaction details (amount, receiver, device, etc.).
2. **Feature Extraction:**
   System derives behavioral features (amount deviation, device mismatch, etc.).
3. **Risk Evaluation:**
    - Rule-based scoring is applied
    - ML model predicts fraud probability
4. **Decision Making:**
   Final risk score is calculated and action is determined (Allow / OTP / Delay / Block).
5. **Result Visualization:**
   User receives decision along with explanation.
6. **Logging & Monitoring:**
   Transaction is stored and displayed in admin dashboard for analysis.

## 5. Design Decisions

- **Hybrid Approach (Rule + ML):**
  Ensures both reliability (rules) and adaptability (ML).
- **Real-Time Processing:**
  Designed to evaluate transactions instantly before completion.
- **Explainability First:**
  Users are informed why decisions are made to build trust.
- **User-Centric Design:**
  Adaptive responses (OTP, delay) reduce unnecessary blocking.
- **Modular Architecture:**
  Enables easy scalability and future upgrades (Kafka, microservices).